

XAVIER FRANKLINE ODHIAMBO

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Summary

- Senior software engineer with 7 years of Python experience building web applications and distributed data systems across SaaS, civic-tech, and procurement platforms. Strong on API engineering (Django REST Framework, FastAPI), database design and optimisation (PostgreSQL, ClickHouse, Elasticsearch), cloud infrastructure (AWS Certified Solutions Architect), and operating distributed systems through CI/CD, testing, observability, and incident response. Recent work includes AI engineering: RAG pipelines, LLM orchestration infrastructure, and self-hosted model deployment, alongside daily AI-assisted development. Open-source contributor to Django and CKAN.

Skills

Languages: Python, JavaScript, TypeScript, SQL, Bash

Backend & APIs: Django, Django REST Framework, FastAPI, Flask, Celery, RabbitMQ, asyncio, RESTful API design

Databases: PostgreSQL, ClickHouse, Elasticsearch, Redis, MongoDB, Alembic

Cloud & Infrastructure: AWS (EC2, RDS, ECS, Lambda, S3, Route 53, SNS), GCP, Docker, Kubernetes, Terraform, Ansible, Linux

CI/CD & Tooling: GitHub Actions, GitLab CI/CD, Bitbucket Pipelines, pytest, Git

Observability: OpenTelemetry, Grafana, Loki, Tempo, Prometheus, Sentry, CloudWatch

AI Engineering & Tools: LlamaIndex, pgvector, OpenRouter, Ollama, OpenWebUI, Claude Code, Cursor

Certifications

AWS Certified Solutions Architect – Amazon Web Services, January 2025

AWS Certified Cloud Practitioner – Amazon Web Services, January 2022

Experience

Senior Software Engineer, Search Atlas (Remote)

August 2025 – June 2026

Backend engineer on an SEO analytics platform integrating Google Search Console, Google Analytics, Google Ads, and Meta Ads for advertising agencies and in-house marketing teams.

- Designed and built REST API services in Django REST Framework with strong input validation, structured error handling, authentication, and rate limiting, backing both internal services and a customer-facing analytics product.
- Built and maintained Django and Celery microservices for third-party API ingestion, asynchronous processing, and scheduled reporting.
- Wrote and shipped PostgreSQL and ClickHouse migrations, optimised slow queries through indexing and schema refactors, and worked on partitioning strategies for high-volume analytics tables.
- Diagnosed and fixed a race condition in scheduled reporting where overlapping Celery Beat batches sent customers duplicate emails, resolving it with Redis-backed distributed locking.
- Owned the email delivery and monitoring component of the platform's reporting microservice, adding AWS SES event tracking for opens, clicks, and bounces and resolving a recurring source of client complaints about undelivered emails.
- Instrumented services with OpenTelemetry across Django and Celery; resolved incidents by correlating Sentry exceptions, Loki logs, and Grafana metrics.
- Used Claude Code daily as a core development workflow: generated CLAUDE.md project context, created architectural and implementation plans from issues ahead of AI-assisted coding, and ran AI-assisted review in the CI pipeline alongside tests and human code review.
- Improved and extended the platform's LLM orchestration layer, adding multi-provider routing, automatic failover, and cost-aware model selection for internal services handling automated analysis and content generation.

Senior Software Engineer, Code for Africa (Remote)

August 2022 – August 2025

Backend and infrastructure engineer building systems for civic-tech and open-data platforms used by researchers, journalists, and policy teams across Africa.

- Built and maintained Django REST Framework, FastAPI, and Flask APIs across multiple platforms, including distributed news indexing on RabbitMQ and Elasticsearch, IoT sensor ingestion (Sensors.Africa), and census and demographic data (Wazimap-NG).
- Lead maintainer of openAFRICA, Code for Africa's CKAN-based open-data portal, and its custom extensions: carried the platform through successive CKAN upgrades from 2.8 to 2.11, including the move to Python 3, and migrated its legacy controllers to Flask blueprints.
- Led the architecture and deployment of multi-node Elasticsearch clusters using Ansible automation, including a Rally-based acceptance benchmark harness that validated indexing throughput, query latency under load, and mixed read/write performance before newly provisioned clusters accepted production traffic.
- Tuned PostgreSQL workloads on Sensors.Africa and Wazimap-NG using query plans and database performance monitoring, resolving N+1 query patterns and adding denormalised and generated fields for hot read paths.

- Implemented end-to-end observability with Grafana, Prometheus, Loki, and Alertmanager, investigating and resolving incidents by correlating Sentry exceptions, logs, and metrics.
- Drove adoption of CI/CD pipelines across engineering teams using GitHub Actions and GitLab CI/CD, with automated testing, containerised image publishing, and technical documentation as standard practice.
- Led development of an internal ad-quality and domain-trust classifier, a gated multi-stage RAG pipeline combining automated signal checks with an LLM-based content-quality stage, gated so the most expensive stage only ran on inputs that had already cleared cheaper checks.
- Led the architecture and operation of a self-hosted LLM platform (Ollama, OpenWebUI) giving non-engineering teams access-controlled use of AI tools for sensitive material that could not go to third-party providers.
- On GCP: provisioned TPU infrastructure for the data team, and deployed Outline VPN for secure internal access.
- Set up and administered MongoDB as the database layer for a PayloadCMS content platform, including collection design and deployment configuration.
- Participated in code review and informally mentored junior engineers and students, sharing technical guidance and feedback.

Software Engineer, QED Solutions Ltd. (Nairobi)

April 2021 – July 2022

Backend engineer on Tendersure Africa, a SaaS B2B procurement platform used across multiple African countries.

- Built and maintained the platform with Django (backend) and Vue.js (frontend), including internationalisation for English, French, Portuguese, and Amharic.
- Integrated localised payment gateways across multiple African markets, handling signed callback verification, idempotent retries, and reconciliation.
- Managed and administered core infrastructure on AWS and DigitalOcean across EC2, RDS, S3, Route 53, and CloudWatch; led implementation of EC2 Auto Scaling groups to improve system availability.
- Led planning and provisioning of infrastructure using Terraform, and migration of legacy applications from VM-based workloads to Docker containers deployed on Amazon ECS.
- Configured AWS CloudWatch for monitoring and SNS for alerting on EC2 and RDS instances, resolving incidents in real time; later introduced a Prometheus and Alertmanager stack for application-level metrics.
- Led adoption of CI/CD pipelines using Bitbucket Pipelines to streamline build, test, and deployment processes.
- Implemented AWS Lambda functions to automate report generation from the core Django application, reducing load on primary application servers.

Software Engineer, Nakala Analytics Ltd. (Nairobi)

August 2020 – March 2021

- Built data analytics applications in Python and Flask with REST APIs over PostgreSQL, integrating data from third-party APIs for downstream visualisation.
- Built ETL pipelines integrating PostgreSQL, dbt for transformation, and Stitch for source replication.
- Architected and deployed JupyterHub using Docker and Docker Swarm for the data analytics team.
- Built frontend interfaces in Vue.js and React for internal data analytics dashboards.

Software Engineer, JobsearchKE (Nairobi)

September 2019 – August 2020

- Built the company's core platform jobsearchke.com from scratch using Django, PostgreSQL, and JavaScript, as the first engineer on the product.
- Architected and deployed application infrastructure on DigitalOcean using Docker, Nginx, and DigitalOcean Spaces.
- Built an automated campaign and marketing platform integrating Mailchimp and SendGrid APIs.

Projects

Algorithmic Trading System, Python

- Built and operate a live algorithmic XAU/USD trading system on a broker platform's Python API, with several strategy configurations running in parallel and automated trade notifications.
- Every strategy change is validated in a custom backtester against historical data before going live, with position and exposure limits enforced in code.

Open Source Contributions

Django: Bug fixes and documentation contributions to the framework.

CKAN: Extension and plugin contributions to the leading open-source platform powering government and institutional open-data portals worldwide.

Education

Jomo Kenyatta University of Agriculture and Technology, BSc Telecommunications Engineering, November 2018